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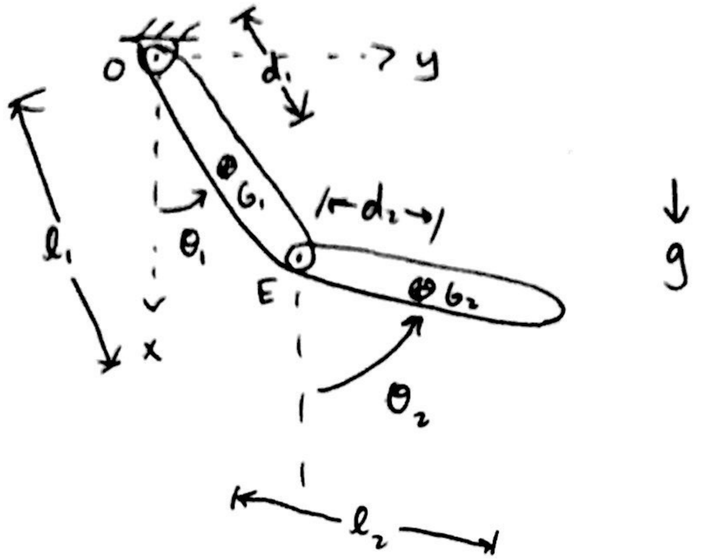
| MAE 4730<br>10/12/2018<br>Lecture 20 | Today:<br>1) Moving support pendulum intuition<br>2) Double pendulum (cont'd) |

## Double Pendulum:

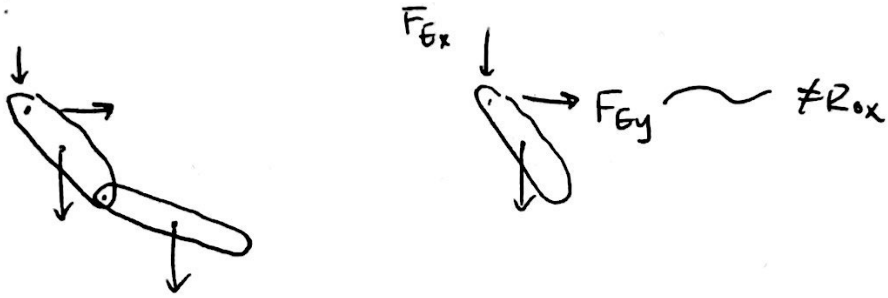



want:

$$\ddot{\theta}_1 = f(\theta_1, \theta_2, \dot{\theta}_1, \dot{\theta}_2, p)$$

$$\ddot{\theta}_2 = f( \qquad )$$

## FBDS:




AMB/o

$$\sum \vec{M}^{ext}_{/o} = \dot{\vec{H}}_{/o}$$

AMB

$$\sum \vec{M}^{ext}_{/E} = \dot{\vec{H}}_{/E}$$

System AMB/o

$$\sum \vec{M}_{/o} = \dot{\vec{H}}_{/o}$$

$$\vec{r}_{G_1/o} \times m_1 g \hat{\imath} + \vec{r}_{G_2/o} \times m_2 g \hat{\imath} = \dot{\vec{H}}_{1/o} + \dot{\vec{H}}_{2/o}$$

$$\vec{r}_{G_1/o} = d \hat{e}_{r_1}$$

$\quad \llcorner\ \hat{e}_{r_1} = \cos\theta_1 \hat{\imath} + \sin\theta_1 \hat{\jmath}$

$$\vec{r}_{G_2/o} = \underbrace{\vec{r}_{E/o}}_{l\hat{e}_{r_1}} + \underbrace{\vec{r}_{G_2/E}}_{d_2 \hat{e}_{r_2}}$$

$\quad \llcorner\ \cos\theta_2 \hat{\imath} + \sin\theta_2 \hat{\jmath}$

$$\dot{\vec{H}}_{1/0} = \vec{r}_{G1/0} \times m_1 \vec{a}_{G1} + I_1^G \ddot{\theta}_1 \hat{k}$$

$$\dot{\vec{H}}_{2/0} = \vec{r}_{G2/0} \times m_2 \vec{a}_{G2} + I_2^G \ddot{\theta}_2 \hat{k}$$

$$\vec{a}_{G1} = \ddot{\theta}_1 d_1 \hat{e}_{\theta_1} - \dot{\theta}_1^2 d_1 \hat{e}_{r_1}$$

$\hookrightarrow \hat{k} \times \hat{e}_{r_1}$

$$\vec{a}_{G2} = \vec{a}_E + \vec{a}_{G2/E}$$

$\hat{k} \times \hat{e}_{r_2}$

$\vec{a}_{G2/E}$: $d_2 \ddot{\theta}_2 \hat{e}_{\theta_2} - \dot{\theta}_1^2 d_2 \hat{e}_{r_2}$

$\vec{a}_E$: $l \ddot{\theta}_1 \hat{e}_{\theta_1} - \dot{\theta}_1^2 l_1 \hat{e}_{r_1}$

Forearm

$AMB_{/E}$

$$\sum \vec{M}^{ext}_{/E} = \dot{\vec{H}}_{/E}$$

$$r_{G2/E} \times m_2 g \hat{\imath} = \vec{r}_{G2/E} \times m_2 \vec{a}_{G2} + I_2^G \ddot{\theta}_2 \hat{k}$$

$\{AMB/_0\}_{\text{system}} \cdot \hat{k}$ = expressions w/ $\ddot{\theta}_1$, $\ddot{\theta}_2$, $\dot{\theta}_1$, $\dot{\theta}_2$, $\theta_1$, $\theta_2$

$\{AMB_{\text{forearm}}/_E\} \cdot \hat{k}$ = ( " )

Solve for $\ddot{\theta}_1$ and $\ddot{\theta}_2$

Note: equations are linear in $\ddot{\theta}_1$ and $\ddot{\theta}_2$

$\Rightarrow$ EOM

$$= \begin{bmatrix} \text{mess} & \text{mess} \\ \text{mess} & \text{mess} \end{bmatrix} \begin{bmatrix} \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{bmatrix} = \begin{bmatrix} \text{mess} \\ \text{mess} \end{bmatrix}$$

messes include $\theta_1$, $\dot{\theta}_1$, $\theta_2$, $\dot{\theta}_2$, parameters